FIN 7082
Group 2: Ava Valley, Gabriela Varisco de Oliveira, Quynh Tran
Group Project - Constructing Equity Factor Fund

I. Overview

A. Fund Name : AGQ Equity Momentum ETF

B. Signals

1. **ROE** - Return on Equity is a measure of financial performance calculated by dividing net income by shareholders' equity. This is also referred to as return on net assets. We choose this factor as a measure of the firm's profitability and how effectively it is utilizing its assets. ROE is an effective comparison tool to other stocks in the same sector. It can serve as an indicator of the future growth rate we are trying to find in the stocks. Extremely high ROE can also be an indicator of risk in a stock where there is little equity in comparison to net income.
2. **R11** - R11 refers to the momentum of the past 12-to-2-month cumulative return. Momentum as a signal brings time relevance to ROE allowing investors to see which stocks have shown an uptrend over the last few months.

C. Objective

1. Large-cap Momentum Factor Fund with an Equity Tilt
2. Large cap companies are stable businesses with notable size and track records. While they are more expensive stocks up front, they usually have experienced management and matured life cycles. We can utilize ROE to see if it can signal to us that a large cap company will continue on its growth trajectory. In times of volatility, large caps have performed better than mid and small caps. This ETF is diversified through many different stocks and benefits from this advantage overtime.

II. Methodology:

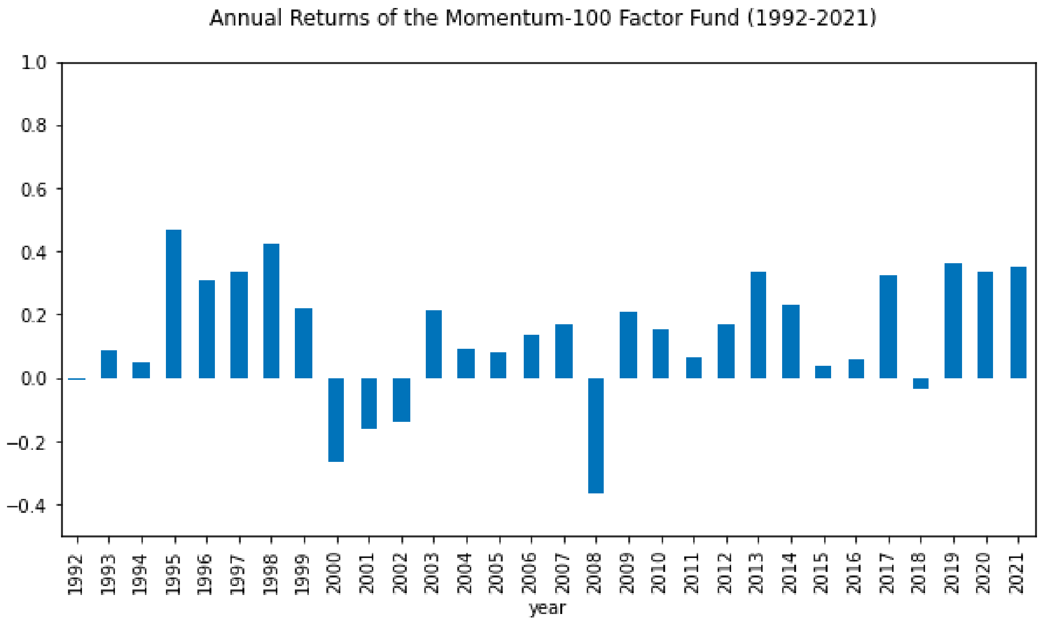
A. Steps of Constructing the Factor Fund

1. *Preparation* - First, we sourced a monthly stock, market capitalization, and beginning-of-month market capitalization files. We formatted date related variables and evaluated the signals including, market capitalization, book-to-market, momentum, investment-to-assets, return on equity, and beta.
2. *Filtering* - After evaluating our variables, it was important to filter to reduce trading costs and volatility and drop stocks with share price below $5 at quarter-end. We wanted to eliminate penny stocks here in order to properly evaluate our results later. We also limited to the top 1000 stocks by market capitalization and trading volume for consistency. We then found the top 100 stocks through our ROE and R11 signals.

3. *Rebalance* - We rebalanced the fund to the year/quarter and eliminated missing returns. Finally, merging our quarter-end sample with monthly returns during the coming quarter. With this data we were able to value-weight the fund portfolio return.
4. *Analysis* - With our results we collected summary statistics as well as created a benchmark index of all eligible stocks for comparison purposes utilizing the same weighting.
5. *Evaluation* - To evaluate our fund, we plotted the time series of its performance, compared to the benchmark, and evaluated using factor funds. Finally, we analyzed the top 10 fund holdings at the end of 2021 to draw our conclusions.

III. Back-Testing

A. Bar plot of annual fund returns over the period from 1992 to 2021.



The bar plot was constructed based on the compounded value weighted annual returns over the period of 1992 - 2021. We computed the compounded VW annual returns, which was represented in the y-axis of the bar plot, by summing up the log returns for each year and then taking the exponential of it. The compounded value weighted annual returns are considered to be a more effective tool to measure the performance of an investment than average returns because with compounded returns compounding effects over researched years and volatility are accounted for in calculations. As we can see from the bar plot, annual returns were positive during the period of 1995-1998, and then dropped during 2000-2002. The returns were getting back to positive results from 2003-2007, and then had a big drop down in 2008. After 2008, the annual returns were recovering back to normal. The drop in 2008 was almost -40% due to the Great Recession. The crash in the stock market was due to panic selling and other underlying economic factors, which led to a big decline in stock price and hence return after one year.

B. Evaluate fund performance by using a simple index of all eligible stocks as the benchmark. Report and discuss at least mean, volatility and the Sharpe ratio of annual returns, as well as the compounded average annual return during the past 1, 3, 5, 10, and 30 years.

- Average annual return of Factor Fund (%): 14.09
- Annual volatility of Factor Fund (%): 20.08
- Annual Sharpe Ratio of Factor Fund: 0.70
- 1-year compounded annual return of Factor Fund (%): 34.93
- 3-year compounded annual return of Factor Fund (%): 34.92
- 5-year compounded annual return of Factor Fund (%): 25.66
- 10-year compounded annual return of Factor Fund (%): 20.79
- 30-year compounded annual return of Factor Fund (%): 12.17

The fund has a mean annual return of 14.09%, volatility of 20.08% and a sharpe ratio of 0.70. This means that if an investor decides to invest in our fund, in one year, his investment will grow at an average of 14%. A volatility of 20% means that the returns may vary around 20% during that one year. Lastly, the sharpe ratio is the average return earned in excess of the risk-free rate per unit of volatility or total risk. A sharpe ratio of 0.7 is sub-optimal, meaning that the risk of our portfolio isn't being offset well enough by its return, ideally we would like to have a sharpe ratio greater than 1. When looking at the 1, 3, 5, 10, and 30 years compounded returns we see that the percentage returns are gradually smaller as the time period increases.

C. Evaluate fund performance using the CAPM and the FF 3-factor model. Report and discuss the factor exposures and risk-adjusted performance of your fund.

CAPM

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                            OLS Regression Results
==============================================================================
Dep. Variable:                 ret_vw   R-squared:                       0.772
Model:                            OLS   Adj. R-squared:                  0.771
Method:                 Least Squares   F-statistic:                     1209.
Date:                Thu, 28 Apr 2022   Prob (F-statistic):          7.83e-117
Time:                        16:51:02   Log-Likelihood:                 865.07
No. Observations:                 360   AIC:                            -1726.
Df Residuals:                     358   BIC:                            -1718.
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.0036      0.001      3.024      0.003       0.001       0.006
MKT            0.9357      0.027     34.775      0.000       0.883       0.989
==============================================================================
Omnibus:                       17.640   Durbin-Watson:                   1.953
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               33.913
Skew:                          -0.265   Prob(JB):                     4.32e-08
Kurtosis:                       4.407   Cond. No.                         23.3
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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The OLS Regression Model / CAPM Model we've built describes the correlation between monthly fund returns and the market excess return. The model only takes correlation of the fund's monthly returns and excess market return into the calculations, so it simply explains the volatility/systematic risk, or also called beta factor of the fund. Beta or coefficient of MKT (excess market return) is calculated as 0.9357, which is considered as a very large number for Beta for a CAPM Model. This can be explained that the fund's return is highly volatile to the changes in market return.

FF3

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                            OLS Regression Results
==============================================================================
Dep. Variable:                 ret_vw   R-squared:                       0.797
Model:                            OLS   Adj. R-squared:                  0.795
Method:                 Least Squares   F-statistic:                     464.7
Date:                Thu, 28 Apr 2022   Prob (F-statistic):          1.04e-122
Time:                        16:51:02   Log-Likelihood:                 885.95
No. Observations:                 360   AIC:                            -1764.
Df Residuals:                     356   BIC:                            -1748.
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      0.0041      0.001      3.667      0.000       0.002       0.006
MKT            0.9351      0.026     35.320      0.000       0.883       0.987
SMB           -0.1340      0.036     -3.744      0.000      -0.204      -0.064
HML           -0.2141      0.035     -6.136      0.000      -0.283      -0.145
==============================================================================
Omnibus:                       20.393   Durbin-Watson:                   2.054
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               50.476
Skew:                          -0.208   Prob(JB):                     1.09e-11
Kurtosis:                       4.787   Cond. No.                         35.8
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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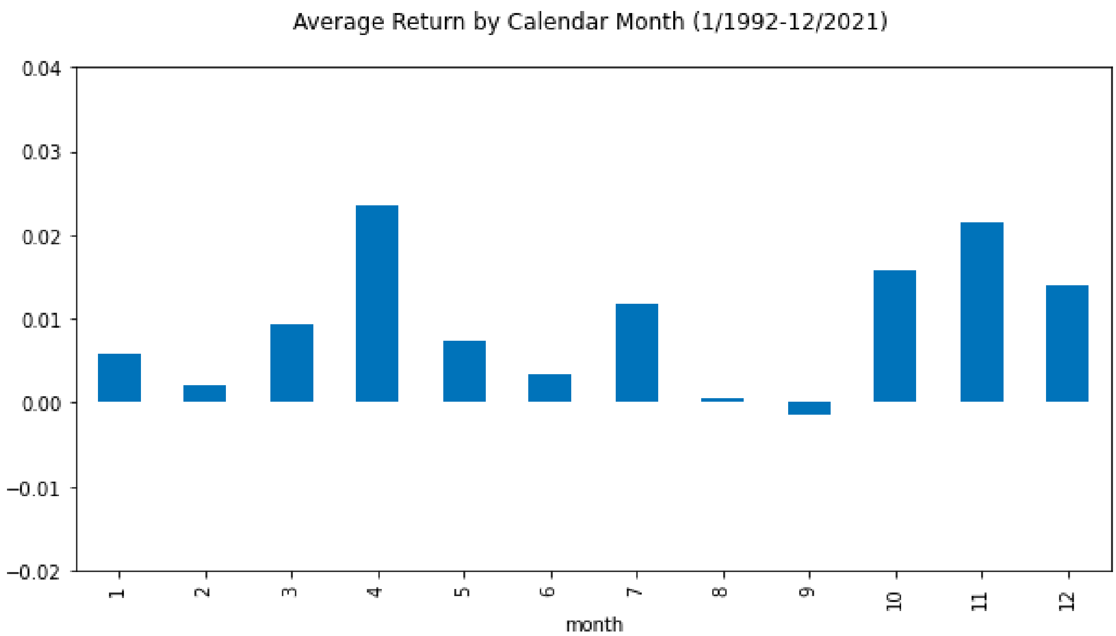
The OSL Regression Model/FF3 we've built above explains correlation between the fund's monthly excess return with three factors: (1) market excess return - MKT, (2) the outperformance of small cap stocks compared to large cap stocks - SMB, and (3) the outperformance of stocks with high book-to-market value compared to stocks with low book-to-market value. Coefficients for MKT, SMB, and HML are 0.9351, -0.1340, and -0.2141, respectively. This indicates that the fund is highly volatile to the market risk premium with correlation coeff of 0.9351. The correlation coeffs with historic excess returns of small cap stocks over large cap stocks and with historic excess returns of high BM stocks over low BM stocks are considerably small and negative (-0.1340 and -0.2141). This indicates that the fund's returns are not positively and largely correlated to and explained by the excess returns of SMB and HML stocks.

D. (4) Any other analysis that you consider informative. I encourage you to be creative here.



We decided to plot the average return by calendar month between the years of 1992 and 2021. That was created by grouping the compounded value weighted annual returns for each month during that period and plotting it with the months on the x-axis and average returns on the y-axis. We can see that the month of April is the one with the highest compounded average return and that September has a negative average return.

IV. Holdings

A. Present a table of the top-10 holdings at the end of 2021 (rank, company name, % fund holding). Comment on fund portfolio concentration and industry exposures.

	permno	MC	rank_R11	wt	ticker
0	14593	2.913284e+07	363	0.101369	AAPL
1	10107	2.525084e+07	149	0.087861	MSFT
2	93436	1.061287e+07	101	0.036928	TSLA
3	14542	9.194035e+06	99	0.031991	GOOG
4	90319	8.714586e+06	103	0.030323	GOOGL
5	13407	7.958976e+06	429	0.027694	FB
6	86580	7.352750e+06	21	0.025584	NVDA
7	92655	4.729411e+06	337	0.016456	UNH
8	47896	4.679664e+06	332	0.016283	JPM
9	66181	4.333696e+06	132	0.015079	HD

Our fund is most concentrated in large cap tech stocks due to their consistent ROE growth. While most of the weight of the top stocks in the fund lies in the tech industry with Apple, Microsoft, Tesla, Google, Facebook, and Nvidia we do have some coverage in other industries as well. In the last 3 spots we see United Healthcare, JP Morgan, and Home Depot. This shows us that the combination of ROE and momentum as signals gives us reliable, growing stocks regardless of industry.

V. Discussions

A. Discuss potential risks that investors should be aware of.

1. This fund is diversified, however drastic events in the tech industry particularly may have an effect on the fund in the short term as it does account for a good portion of stocks.
2. Though our fund manages fees and costs by limiting transactions, ETFs are not purely passive products and some of this cost will be incurred by our holders.
3. Due to our fund being equity and momentum based, but large cap there may be low volatility. A holder’s equity will not drastically shoot up on a given day.

B. Other concerns and limitations of your fund.

1. Large cap ETFs may not have the ability for price appreciation because they have already been discovered by the market.
2. There is still opportunity for a better mix of growth and value stocks to be able to hold the performance when value slumps.